Übungsaufgaben: Gleichungen

Löse die Aufgaben in deinem Heft. ★☆☆ einfach · ★★☆ mittel · ★★★ schwer. Die Lösungen stehen auf Seite 2.

Name: _____ Datum: _____

★☆☆ Einfach — Grundwissen

1. $-8x - 12 = -60$

2. $7x - 6 = 1$

3. $5x + 5 = 35$

4. $2x + 5 = -5$

5. $-x + 2 = -5$

6. $3x + 7 = 10$

7. $-2x + 9 = 19$

8. $6x + 4 = -44$

9. $-8x - 11 = 29$

10. $6x + 9 = 51$

11. $8x - 12 = -60$

12. $-2x + 1 = 7$

13. $-5x - 4 = 26$

14. $-6x + 5 = 41$

15. $-7x + 5 = -30$

16. $-9x + 12 = 57$

17. $6x + 10 = -44$

18. $-5x + 7 = -33$

19. $-3x - 4 = 8$

20. $-4x - 1 = -5$

21. $8x + 3 = -37$

22. $4x + 5 = -11$

★★☆ Mittel — Anwenden

23. $8x + 2 = -4x + 62$

24. $-x + 2 = 5x + 14$

25. $4x + 8 = -5x + 62$

26. $-7x + 1 = -x + 31$

27. $-4x + 9 = -5x + 12$

28. $4x - 9 = 3x - 14$

29. $7x - 1 = 41$

30. $2x = -4x + 12$

31. $2x + 6 = -x + 3$

32. $-2x - 6 = x - 27$

33. $7x + 3 = 24$

34. $4x = 2x + 14$

35. $-6x - 7 = -5x + 1$

36. $5x - 7 = 3x - 3$

37. $8x + 8 = -2x + 88$

38. $4x + 4 = 2x - 10$

39. $5x = 4x + 2$

40. $4x - 7 = -5x - 52$

41. $4x + 8 = 5x + 4$

42. $3x + 3 = 2x + 5$

43. $-4x + 5 = 3x + 61$

44. $2x - 6 = -4x - 30$

★★★ Schwer — Knobeln und Transfer

45. $4(2x - 5) = 2x - 23$

46. $\frac{x}{3} - 4 = 1$

47. $(x + 3)(x + 3) = (x + 2)(x - 5) + 55$

48. $-3(3x - 3) = -2x + 30$

49. $\frac{x}{2} + 5 = 9$

50. $(x + 4)(x + 3) = (x + 1)(x - 3) + 24$

51. $3(x + 3) = -x + 17$

52. $\frac{x}{5} - 3 = 0$

53. $(x + 3)(x + 3) = (x + 4)(x - 5) + 57$

54. $5(x + 4) = x + 12$

55. $\frac{x}{5} + 6 = 4$

56. $(x + 1)(x - 2) = (x + 1)(x + 1) - 6$

57. $3(3x - 2) = -2x - 50$

58. $\frac{x}{3} - 2 = -5$

59. $(x + 3)(x - 3) = (x + 3)(x + 1) - 36$

60. $6(x + 2) = 3x + 18$

61. $\frac{x}{2} + 2 = 3$

62. $(x + 3)(x + 3) = (x + 2)(x + 1) - 8$

63. $-3(x + 5) = 2x - 40$

64. $\frac{x}{2} - 4 = -10$

65. $(x + 1)(x - 4) = (x + 2)(x + 1) + 18$

Lösungen: Gleichungen

Lösungen · Seite 2

Vergleiche mit deinen Ergebnissen. Die Nummern entsprechen den Aufgaben auf Seite 1.

★☆☆ Einfach — Grundwissen

1. $-8x = -48 \Rightarrow \boldsymbol{x = 6}$

2. $7x = 7 \Rightarrow \boldsymbol{x = 1}$

3. $5x = 30 \Rightarrow \boldsymbol{x = 6}$

4. $2x = -10 \Rightarrow \boldsymbol{x = -5}$

5. $-x = -7 \Rightarrow \boldsymbol{x = 7}$

6. $3x = 3 \Rightarrow \boldsymbol{x = 1}$

7. $-2x = 10 \Rightarrow \boldsymbol{x = -5}$

8. $6x = -48 \Rightarrow \boldsymbol{x = -8}$

9. $-8x = 40 \Rightarrow \boldsymbol{x = -5}$

10. $6x = 42 \Rightarrow \boldsymbol{x = 7}$

11. $8x = -48 \Rightarrow \boldsymbol{x = -6}$

12. $-2x = 6 \Rightarrow \boldsymbol{x = -3}$

13. $-5x = 30 \Rightarrow \boldsymbol{x = -6}$

14. $-6x = 36 \Rightarrow \boldsymbol{x = -6}$

15. $-7x = -35 \Rightarrow \boldsymbol{x = 5}$

16. $-9x = 45 \Rightarrow \boldsymbol{x = -5}$

17. $6x = -54 \Rightarrow \boldsymbol{x = -9}$

18. $-5x = -40 \Rightarrow \boldsymbol{x = 8}$

19. $-3x = 12 \Rightarrow \boldsymbol{x = -4}$

20. $-4x = -4 \Rightarrow \boldsymbol{x = 1}$

21. $8x = -40 \Rightarrow \boldsymbol{x = -5}$

22. $4x = -16 \Rightarrow \boldsymbol{x = -4}$

★★☆ Mittel — Anwenden

23. $12x = 60 \Rightarrow \boldsymbol{x = 5}$

24. $-6x = 12 \Rightarrow \boldsymbol{x = -2}$

25. $9x = 54 \Rightarrow \boldsymbol{x = 6}$

26. $-6x = 30 \Rightarrow \boldsymbol{x = -5}$

27. $x = 3 \Rightarrow \boldsymbol{x = 3}$

28. $x = -5 \Rightarrow \boldsymbol{x = -5}$

29. $7x = 42 \Rightarrow \boldsymbol{x = 6}$

30. $6x = 12 \Rightarrow \boldsymbol{x = 2}$

31. $3x = -3 \Rightarrow \boldsymbol{x = -1}$

32. $-3x = -21 \Rightarrow \boldsymbol{x = 7}$

33. $7x = 21 \Rightarrow \boldsymbol{x = 3}$

34. $2x = 14 \Rightarrow \boldsymbol{x = 7}$

35. $-x = 8 \Rightarrow \boldsymbol{x = -8}$

36. $2x = 4 \Rightarrow \boldsymbol{x = 2}$

37. $10x = 80 \Rightarrow \boldsymbol{x = 8}$

38. $2x = -14 \Rightarrow \boldsymbol{x = -7}$

39. $x = 2 \Rightarrow \boldsymbol{x = 2}$

40. $9x = -45 \Rightarrow \boldsymbol{x = -5}$

41. $-x = -4 \Rightarrow \boldsymbol{x = 4}$

42. $x = 2 \Rightarrow \boldsymbol{x = 2}$

43. $-7x = 56 \Rightarrow \boldsymbol{x = -8}$

44. $6x = -24 \Rightarrow \boldsymbol{x = -4}$

★★★ Schwer — Knobeln und Transfer

45. $8x - 20 = 2x - 23 \Rightarrow \boldsymbol{x = -\frac{1}{2}}$

46. $\frac{x}{3} = 5 \Rightarrow \boldsymbol{x = 15}$

47. x^2 fällt weg: $6x + 9 = -3x + 45 \Rightarrow \boldsymbol{x = 4}$

48. $-9x + 9 = -2x + 30 \Rightarrow \boldsymbol{x = -3}$

49. $\frac{x}{2} = 4 \Rightarrow \boldsymbol{x = 8}$

50. x^2 fällt weg: $7x + 12 = -2x + 21 \Rightarrow \boldsymbol{x = 1}$

51. $3x + 9 = -x + 17 \Rightarrow \boldsymbol{x = 2}$

52. $\frac{x}{5} = 3 \Rightarrow \boldsymbol{x = 15}$

53. x^2 fällt weg: $6x + 9 = -x + 37 \Rightarrow \boldsymbol{x = 4}$

54. $5x + 20 = x + 12 \Rightarrow \boldsymbol{x = -2}$

55. $\frac{x}{5} = -2 \Rightarrow \boldsymbol{x = -10}$

56. x^2 fällt weg: $-x - 2 = 2x - 5 \Rightarrow \boldsymbol{x = 1}$

57. $9x - 6 = -2x - 50 \Rightarrow \boldsymbol{x = -4}$

58. $\frac{x}{3} = -3 \Rightarrow \boldsymbol{x = -9}$

59. x^2 fällt weg: $-9 = 4x - 33 \Rightarrow \boldsymbol{x = 6}$

60. $6x + 12 = 3x + 18 \Rightarrow \boldsymbol{x = 2}$

61. $\frac{x}{2} = 1 \Rightarrow \boldsymbol{x = 2}$

62. x^2 fällt weg: $6x + 9 = 3x - 6 \Rightarrow \boldsymbol{x = -5}$

63. $-3x - 15 = 2x - 40 \Rightarrow \boldsymbol{x = 5}$

64. $\frac{x}{2} = -6 \Rightarrow \boldsymbol{x = -12}$

65. x^2 fällt weg: $-3x - 4 = 3x + 20 \Rightarrow \boldsymbol{x = -4}$